Kyra Baranick

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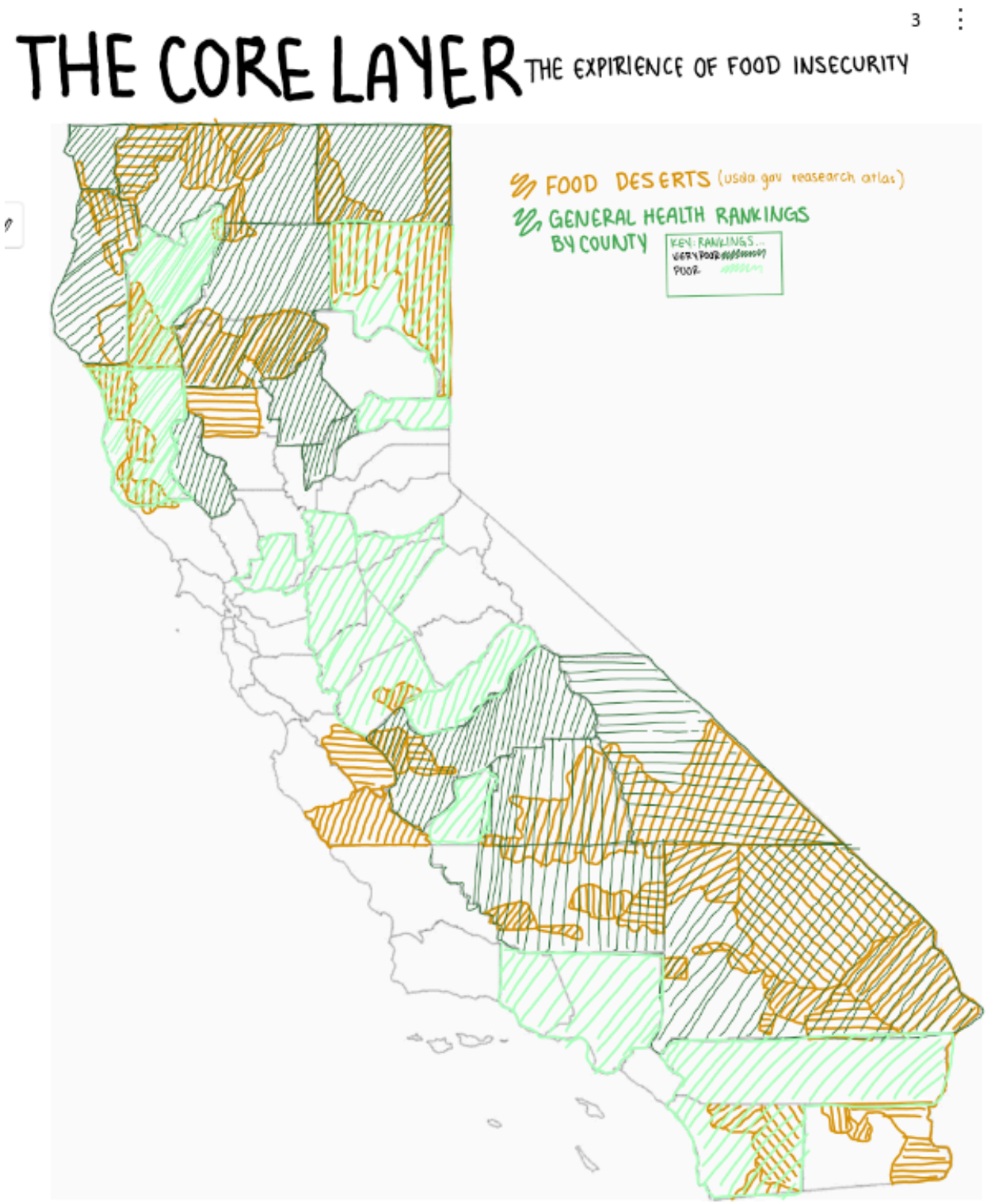
# Mapping the Layers of Food Insecurity: Artistic Project

In a state where population is high and resources are scarce it can be challenging to provide adequate healthy food to all residents, leaving many with poor overall health and resource inaccessibility. Food deserts, specifically, are defined as areas with limited access to supermarkets and healthy food sources. As food access is essential for maintaining health the following maps aim to explore the relationship between food deserts in California and other geographic factors. By overlaying maps of factors such as median income, water scarcity and overall health we can better examine how these factors interact and influence systematic access to nutritious food. Systematic access or lack of access is referred to as “food apartheid” meaning even if our food system contains an abundance of great nutritious food there are people excluded from that abundance (Union of Concerned Sciences). These inequities will not just “go away” so in order to attack this injustice head on we must understand where the injustice is occurring and the primary factors that contribute to it.




(County Health Rankings & Roadmaps. (n.d.))

GENERAL HEALTH AND FOOD INSECURITY

Before we examine the factors causing food insecurity it is important to explore the broader experiences of those in “food deserted” counties. Many food dessert areas rely heavily on fast food for their daily source of nutrition. Lack of nutritious food is a direct contributor to overall health. Consuming processed food over an extended period can lead to obesity, hypertension, type 2 diabetes, and cardiovascular diseases, all of which are observed at high rates in communities affected by food injustice (National Academy of Sciences). A 2016 UCS analysis regarding specifically diabetes, found that increasing the number of fresh food sources would significantly lower the rates of diabetes in particular (Union of Unconcerned Sciences)When mapping the relationship between food insecurity and general health rankings we can visually observe a correlation. The majority of the food deserts (in orange) overlap with the most severe and second most severe lack of overall health in California. Studies show this is no coincidence proving that areas with food deserts consume more fast food and especially more soft drinks (National Center for Biotechnology Information. (n.d.)).

# SOCIAL FACTORS LOW INCOME AREAS

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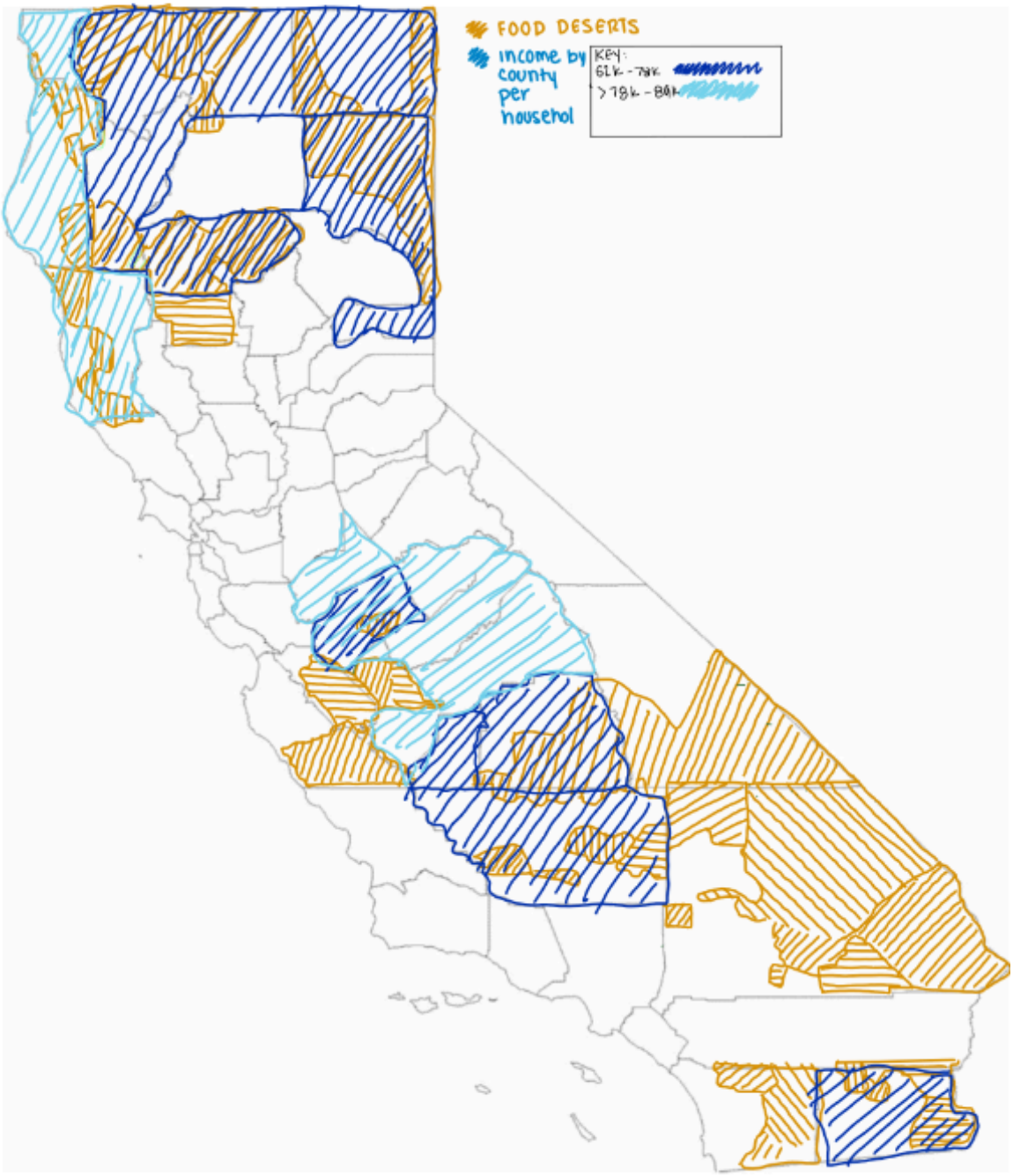


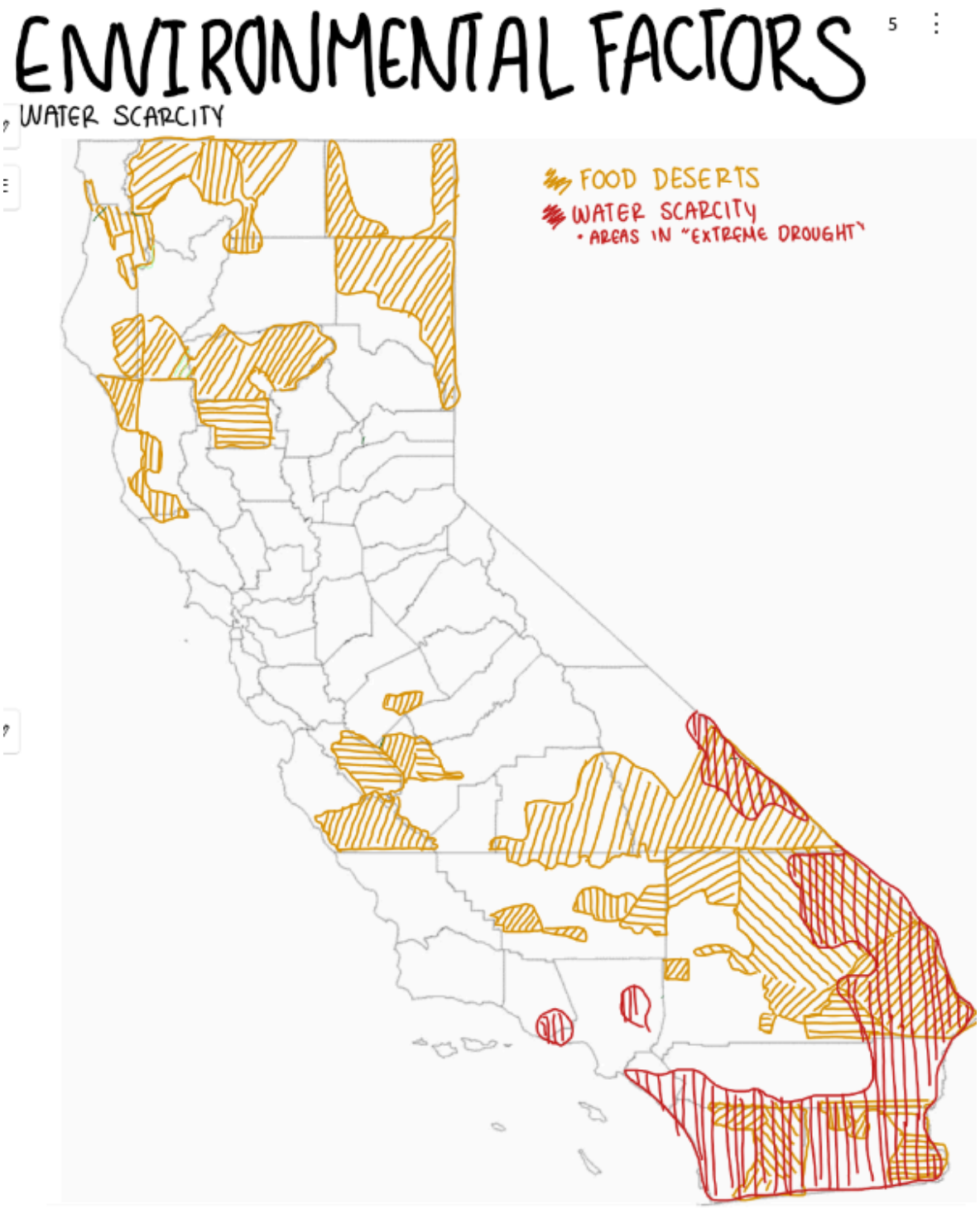

(GeoCurrents. (n.d.).)

MEDIAN INCOME BY HOUSHOLD

It is important to note that food accessibility does not just come from building new grocery stores, as access to healthy food requires two components. Firstly, the presence of the food and secondly, the ability of the residents to access the food. In other words, there is no use having healthy options if the residents cannot afford them. For example, a study by the national research council has concluded that the consumption of soft drinks has doubled since 1965 and perhaps not coincidentally the relative price has dropped as well (National Center for Biotechnology Information. (n.d.)). In some counties it would be effective to bring in some more healthy beverage options such as fruit juice or kombucha however these can be expensive so it is important to look at income when attempting to devise solutions.

As seen in the map, low income areas tend to follow the same trends as food injustice does and these areas also tend to contain minority communities and communities of color. A study by NCBI proved people of color are more impacted by food injustice and therefore more susceptible to diet related illness (National Center for Biotechnology Information. (n.d.)). . The root of this issue comes ultimately from systemic injustices starting in income. According to the Federal Reserve Board, “Black families' median wealth is less than 15 percent that of white families ($44,900 vs. $285,000) and the median wealth of Latino families is about 20 percent that of white families ($61,000 vs. $285,000) (Feeding America). Aside from income disparity, an effective way to disempower a community is to give them low levels of representation. Black, latinx, and Native Americans are significantly underrepresented when it comes to

the farming industry (Union of Concerned Sciences). So not only are there inequities in income there are contributions to food injustice involving fields of work  and representation within those fields.  Everyone deserves the means to access food and these communities are currently being discriminated against by persistent racial gaps that are rooted in a history of structurally systemic racism that contribute to the food insecurity disparities we see today (Feeding America).




(NBC  Los Angeles)

WATER SCARCITY

Because California is a highly populated state who experiences frequent droughts, water remains a crucial topic of discussion. When mapping the relationship between food insecurity and extreme water scarcity we can see that the areas with extreme food deserts and extreme water scarcity overlap in the south part of the state but not so much in the north. It is especially a challenge to get water to the southern half of the state because it is the most densely populated and receives the least amount of rainfall. Water in the southern half of the state comes from the delta watershed but mostly form the wetter north and central parts of the state (Barraza). This makes water in scarce regions very expensive as it requires a lot of resources and energy to transport. Clean water is a basic right of every citizen and this scarcity will directly affect the water accessibility of citizens in the extremely water scarce areas. Perhaps more indirectly inaccessibility in water can lead to inaccessibility in food. Regions with extreme water scarcity do not grow many crops, meaning there are not many local farmers or farmers markets. Even if a citizen took the initiative to grow their own sustainable food it would be more challenging in these areas with the dry climate and shortage of water.  A possible solution could be to incentivise farmers to sell their “misfit” or non aesthetically pleasing (but still quality) products to these areas at a discounted rate.

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